SMART MOBILITY

CHARGEOPS

OBJECTIVE

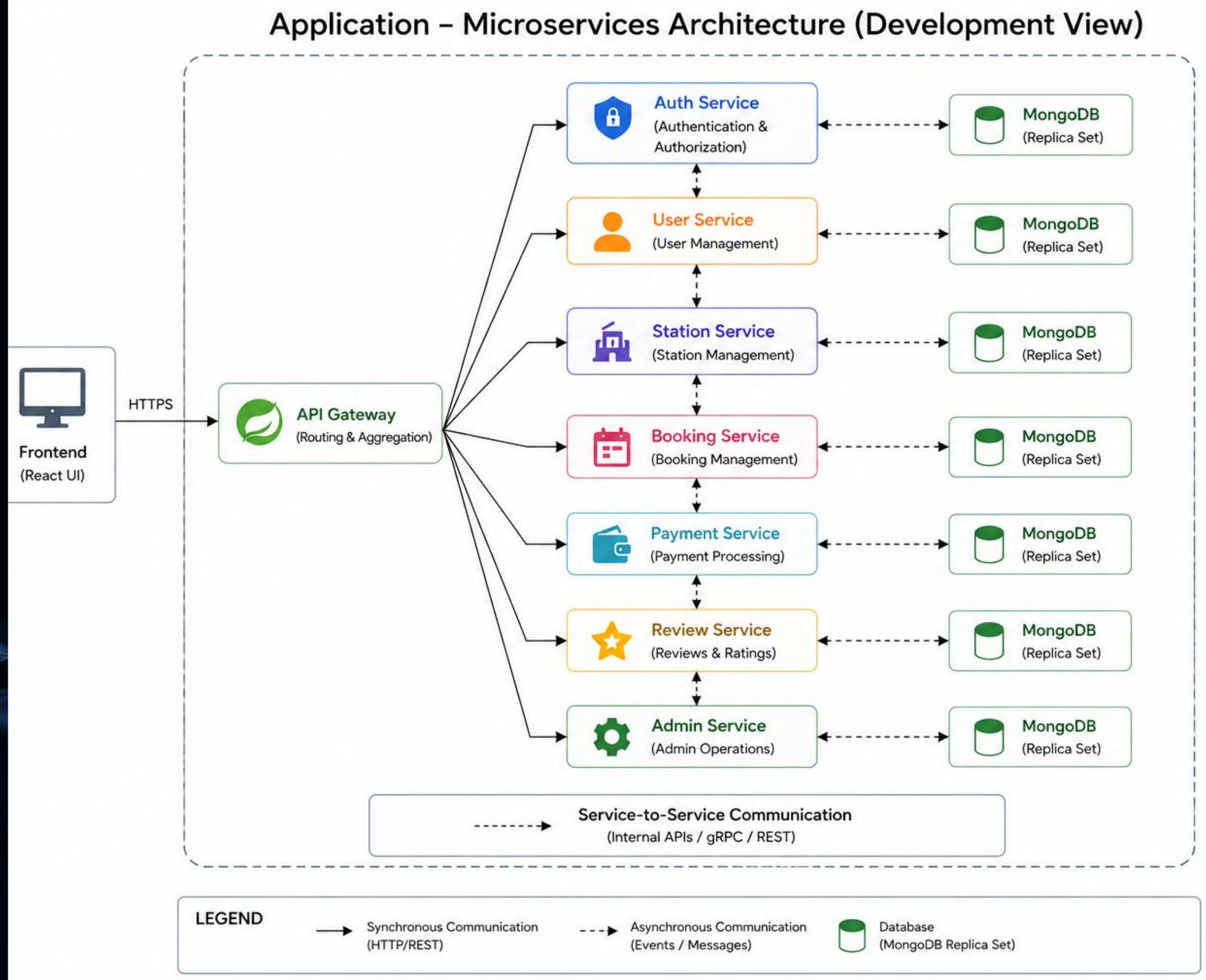
- Build EV charging slot registration system
- Automate build, test, security scan, deployment
- Ensure secure and scalable Kubernetes deployment
- Implement DevSecOps practices

LARANA,
INC.

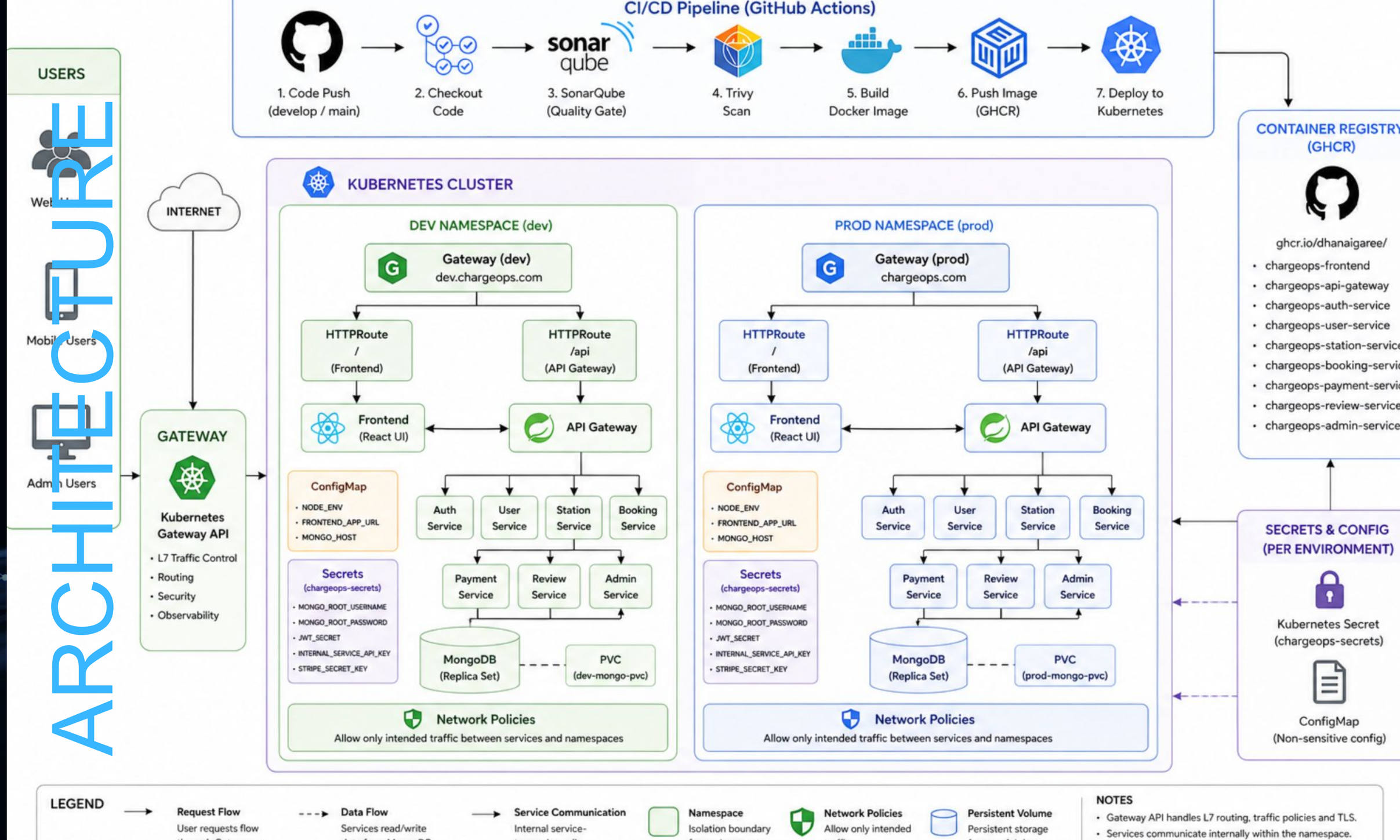
OVERVIEW

- User books EV charging slots
- Frontend UI: Slot booking interface
- Backend API: Handles booking logic
- Database: Stores users, slots, reservations

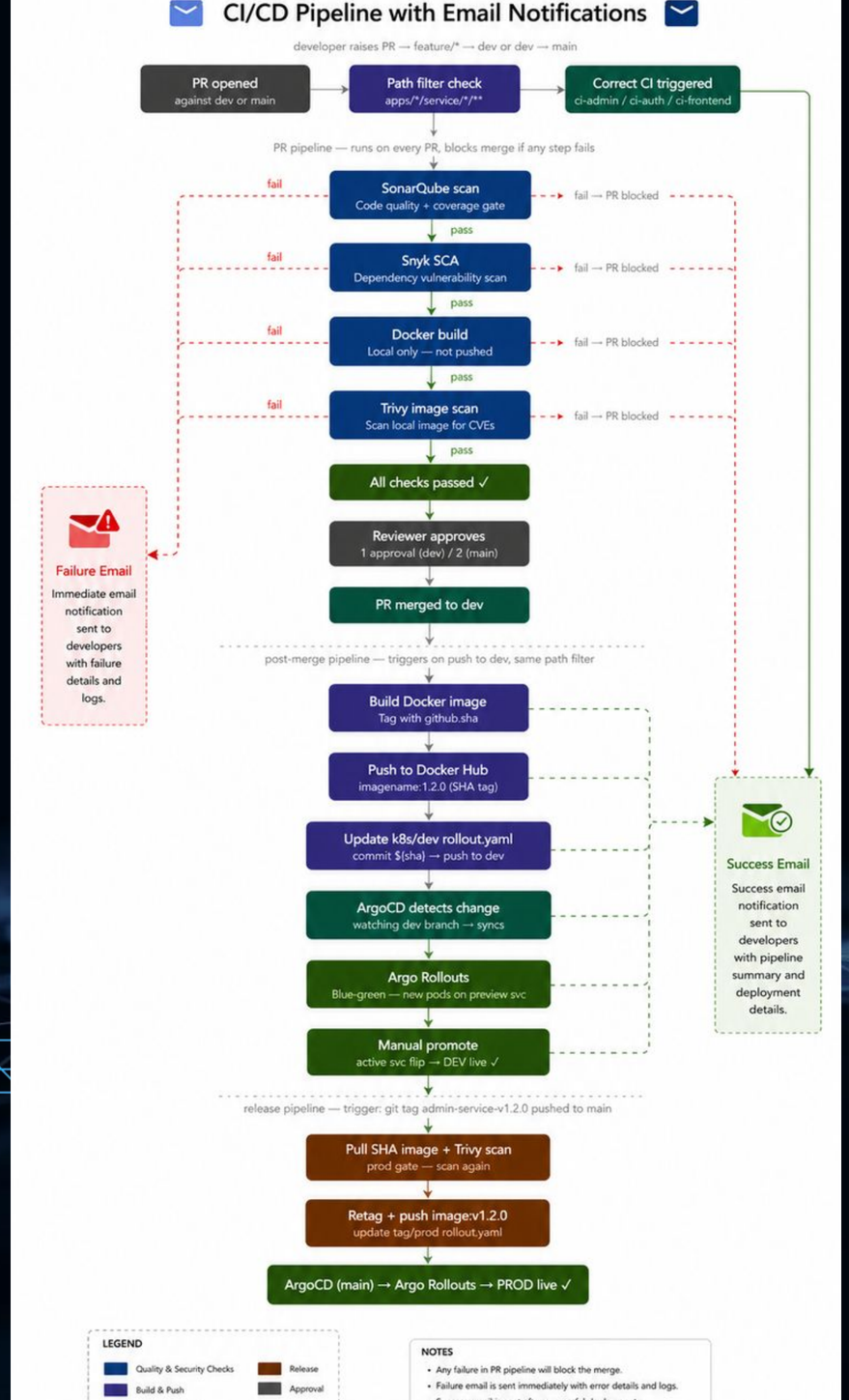
APPLICATION ARCHITECTURE



DEVOPS ARCHITECTURE



CI/CD PIPELINE FLOW



DOCKER

Docker is a containerization platform used to package applications and their dependencies into lightweight, portable containers.

```
-31-5-27:/home/ubuntu/ChargeOps# docker ps
D    IMAGE                        COMMAND                    CREATED          STATUS          PORTS
d    ev-backend-frontend          "/docker-entrypoint.…"     30 hours ago     Up 8 hours      80/tcp
1    ev-backend-api-gateway       "docker-entrypoint.s…"     30 hours ago     Up 8 hours      0.0.0.
2    ev-backend-admin-service     "docker-entrypoint.s…"     30 hours ago     Up 8 hours      0.0.0.
7    ev-backend-payment-service   "docker-entrypoint.s…"     30 hours ago     Up 8 hours      0.0.0
6    ev-backend-review-service    "docker-entrypoint.s…"     30 hours ago     Up 8 hours      0.0.0.
4    ev-backend-auth-service      "docker-entrypoint.s…"     30 hours ago     Up 8 hours      0.0.0.
6    ev-backend-booking-service   "docker-entrypoint.s…"     30 hours ago     Up 8 hours      0.0.0
1    ev-backend-station-service   "docker-entrypoint.s…"     30 hours ago     Up 8 hours      0.0.0
c    ev-backend-user-service      "docker-entrypoint.s…"     30 hours ago     Up 8 hours      0.0.0.
b    mongo:7                      "docker-entrypoint.s…"     30 hours ago     Up 8 hours      0.0.0
3    mongo:7                      "docker-entrypoint.s…"     30 hours ago     Up 8 hours      0.0.0
4    mongo:7                      "docker-entrypoint.s…"     30 hours ago     Up 8 hours      0.0.0
4    mongo:7                      "docker-entrypoint.s…"     30 hours ago     Up 8 hours      0.0.0
```

KUBERNETES

DEV

```
root@ip-172-31-5-27:/home/ubuntu/ChargeOps# kubectl get pods -n dev
NAME                                  READY   STATUS    RESTARTS      AGE
admin-service-667655bc49-bbh4g        1/1     Running   2 (28m ago)   11h
api-gateway-78d55f6895-mzskc          1/1     Running   1 (28m ago)   11h
auth-service-6b486cd977-vkjxm         1/1     Running   2 (28m ago)   11h
booking-service-f8875cc77-b784b       1/1     Running   2 (28m ago)   11h
frontend-5cf86f8559-thcdn             1/1     Running   0             72s
mongodb-5d5b87fcf5-vkgfh              1/1     Running   1 (28m ago)   11h
payment-service-57c87c4667-rnrvs      1/1     Running   2 (28m ago)   11h
review-service-549db8dd49-p7dtj       1/1     Running   2 (28m ago)   11h
station-service-594467dd64-npvfk      1/1     Running   2 (28m ago)   11h
user-service-995cd8bb5-7qr9p          1/1     Running   2 (28m ago)   11h
root@ip-172-31-5-27:/home/ubuntu/ChargeOps#
```

PROD

```
^Croot@ip-172-31-5-27:/home/ubuntu/ChargeOps# kubectl get pods -n prod
NAME                                  READY   STATUS    RESTARTS   AGE
admin-service-7db64f4946-c9kcm        1/1     Running   0          44s
api-gateway-f9779f48b-jgs8d           1/1     Running   0          96s
auth-service-5db8fbf9b7-pdjp4         1/1     Running   0          50s
booking-service-77564bcbd7-mzn8g      1/1     Running   0          45s
frontend-7948468c87-zlttz             1/1     Running   0          21m
mongodb-7547767ffb-fnhrl              1/1     Running   0          21m
payment-service-7df8dd6bb6-9d7vn      1/1     Running   0          50s
review-service-b95c8757b-wrxsj        1/1     Running   0          58s
station-service-569f56b5bb-mc2rp      1/1     Running   0          56s
user-service-97f6764c9-s5286          1/1     Running   0          61s
```

DEV

```
root@ip-172-31-5-27:/home/ubuntu/ChargeOps# kubectl get all -n dev
NAME                                      READY   STATUS    RESTARTS
pod/admin-service-667655bc49-bbh4g        1/1     Running   2 (28m ago)
pod/api-gateway-78d55f6895-mzskc          1/1     Running   1 (29m ago)
pod/auth-service-6b486cd977-vkjxm         1/1     Running   2 (29m ago)
pod/booking-service-f8875cc77-b784b       1/1     Running   2 (29m ago)
pod/frontend-5cf86f8559-thcdn             1/1     Running   0
pod/mongodb-5d5b87fcf5-vkgfh              1/1     Running   1 (29m ago)
pod/payment-service-57c87c4667-rnrvs      1/1     Running   2 (28m ago)
pod/review-service-549db8dd49-p7dtj       1/1     Running   2 (28m ago)
pod/station-service-594467dd64-npvfk      1/1     Running   2 (29m ago)
pod/user-service-995cd8bb5-7qr9p          1/1     Running   2 (28m ago)

NAME                      TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
service/admin-service     ClusterIP   10.106.104.206   <none>        8007/TCP       13h
service/api-gateway       ClusterIP   10.103.170.131   <none>        8000/TCP       13h
service/auth-service      ClusterIP   10.97.92.76      <none>        8001/TCP       13h
service/booking-service   ClusterIP   10.107.178.12    <none>        8004/TCP       13h
service/frontend          NodePort    10.104.172.130   <none>        80:32176/TCP   13h
service/mongodb           ClusterIP   10.107.80.177    <none>        27017/TCP      13h
service/payment-service   ClusterIP   10.98.161.214    <none>        8005/TCP       13h
service/review-service    ClusterIP   10.111.97.75     <none>        8006/TCP       13h
service/station-service   ClusterIP   10.103.92.240    <none>        8003/TCP       13h
service/user-service      ClusterIP   10.107.67.199    <none>        8002/TCP       13h
```

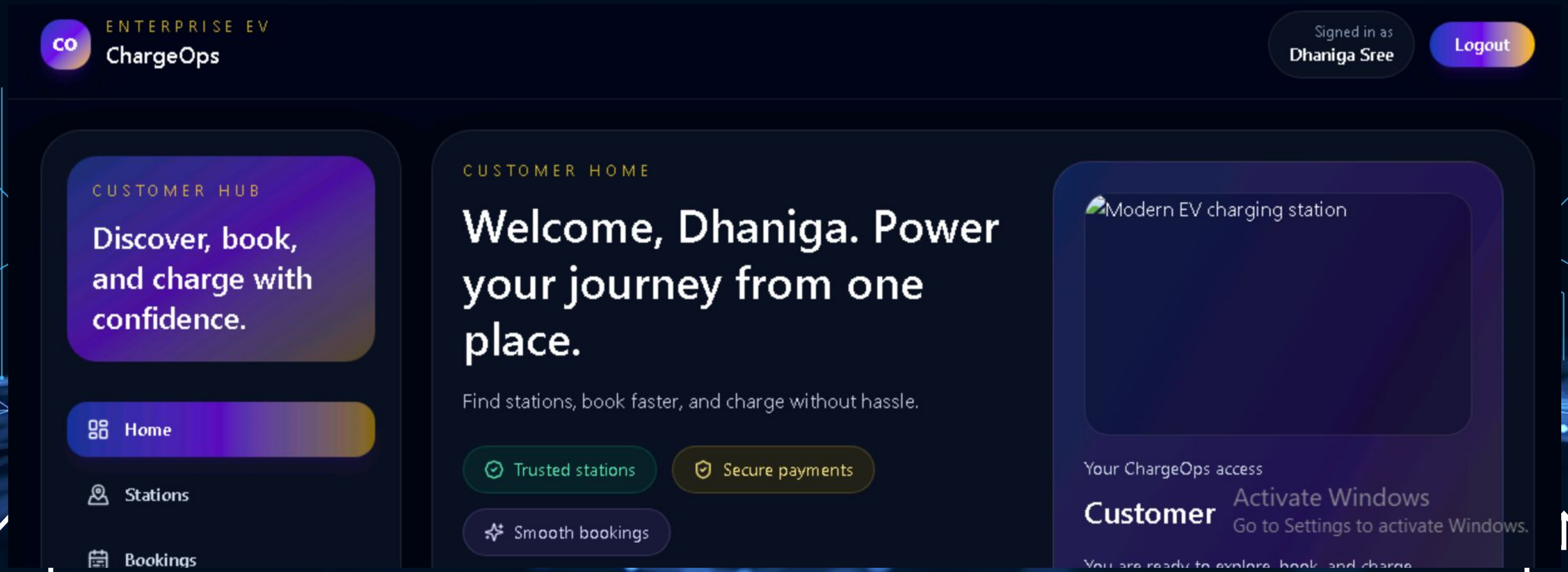
```
root@ip-172-31-5-27:/home/ubuntu/ChargeOps# kubectl get all -n dev
NAME                                      READY   STATUS    RESTARTS      AGE
pod/admin-service-667655bc49-bbh4g        1/1     Running   2 (28m ago)   11h
pod/api-gateway-78d55f6895-mzskc          1/1     Running   1 (29m ago)   11h
pod/auth-service-6b486cd977-vkjxm         1/1     Running   2 (29m ago)   11h
pod/booking-service-f8875cc77-b784b       1/1     Running   2 (29m ago)   11h
pod/frontend-5cf86f8559-thcdn             1/1     Running   0             113s
pod/mongodb-5d5b87fcf5-vkgfh              1/1     Running   1 (29m ago)   11h
pod/payment-service-57c87c4667-rnrvs      1/1     Running   2 (28m ago)   11h
pod/review-service-549db8dd49-p7dtj       1/1     Running   2 (28m ago)   11h
pod/station-service-594467dd64-npvfk      1/1     Running   2 (29m ago)   11h
pod/user-service-995cd8bb5-7qr9p          1/1     Running   2 (28m ago)   11h

NAME                      TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
service/admin-service     ClusterIP   10.106.104.206   <none>        8007/TCP       13h
service/api-gateway       ClusterIP   10.103.170.131   <none>        8000/TCP       13h
service/auth-service      ClusterIP   10.97.92.76      <none>        8001/TCP       13h
service/booking-service   ClusterIP   10.107.178.12    <none>        8004/TCP       13h
service/frontend          NodePort    10.104.172.130   <none>        80:32176/TCP   13h
service/mongodb           ClusterIP   10.107.80.177    <none>        27017/TCP      13h
service/payment-service   ClusterIP   10.98.161.214    <none>        8005/TCP       13h
service/review-service    ClusterIP   10.111.97.75     <none>        8006/TCP       13h
service/station-service   ClusterIP   10.103.92.240    <none>        8003/TCP       13h
service/user-service      ClusterIP   10.107.67.199    <none>        8002/TCP       13h

NAME                                 READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/admin-service        1/1     1            1           13h
deployment.apps/api-gateway          1/1     1            1           13h
deployment.apps/auth-service         1/1     1            1           13h
deployment.apps/booking-service      1/1     1            1           13h
deployment.apps/frontend             1/1     1            1           13h
deployment.apps/mongodb              1/1     1            1           13h
deployment.apps/payment-service      1/1     1            1           13h
deployment.apps/review-service       1/1     1            1           13h
deployment.apps/station-service      1/1     1            1           13h
deployment.apps/user-service         1/1     1            1           13h
```

KUBECTL GET ALL

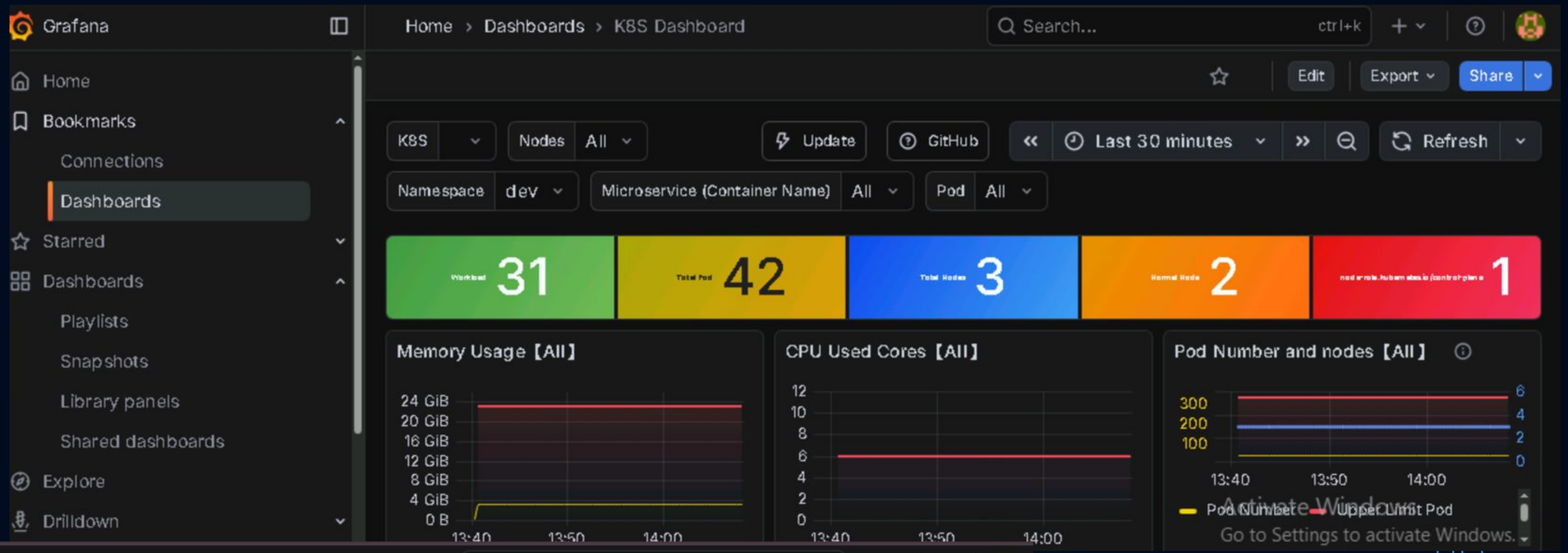
KUBECTL GET

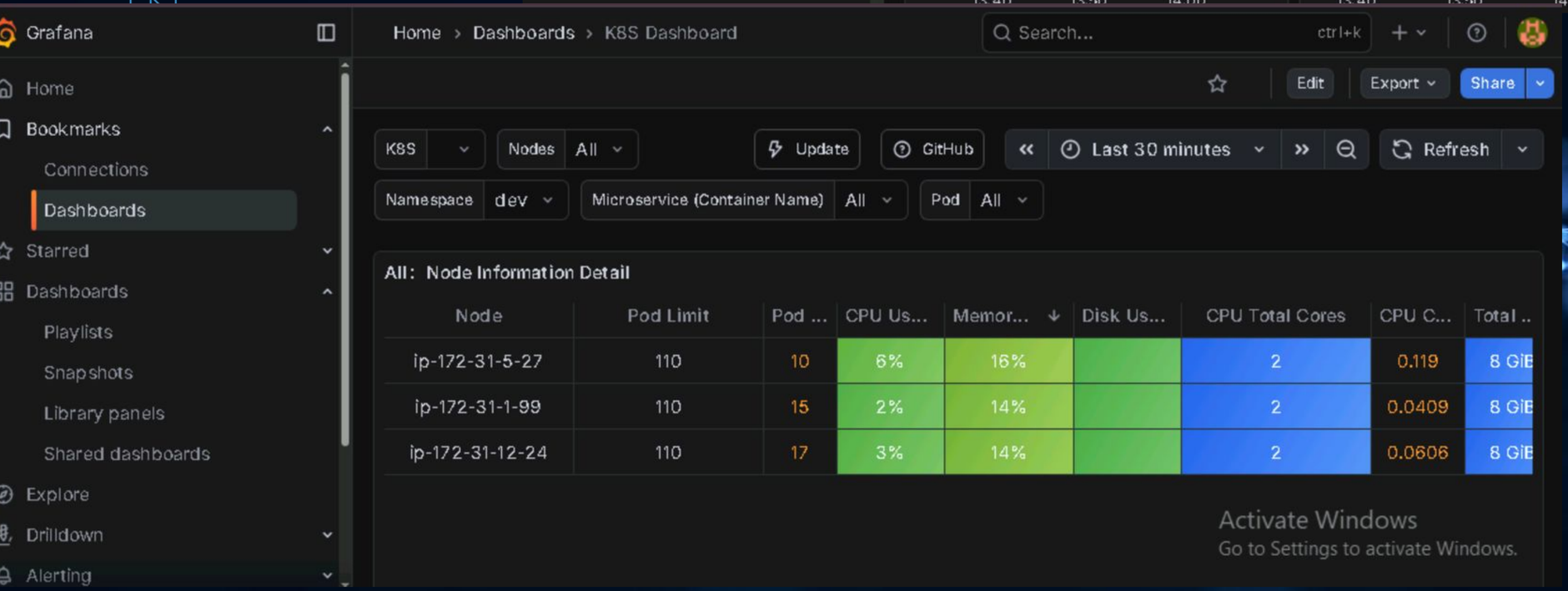
```
root@ip-172-31-5-27:/home/ubuntu/ChargeOps# kubectl get all -n prod
NAME                                    READY   STATUS    RESTARTS        AGE
pod/admin-service-5f8586984c-2hshw      0/1     Running   1 (118s ago)    7h4m
pod/auth-service-775ffc9b95-n44dv       0/1     Running   1 (117s ago)    7h4m
pod/booking-service-5dddf759c9-4c7lj    0/1     Running   1 (117s ago)    7h4m
pod/chargeops-66c54854f4-b2kzf          1/1     Running   1 (117s ago)    7h4m
pod/frontend-68b6f947bc-c6glh           1/1     Running   1 (117s ago)    7h4m
pod/mongodb-0                           1/1     Running   1 (117s ago)    7h22m
pod/payment-service-868bb66dd7-7nqw2    0/1     Running   1 (118s ago)    7h4m
pod/review-service-9f5df9d6c-82k6t      0/1     Running   1 (118s ago)    7h4m
pod/station-service-79b9b94f66-b7t7z    0/1     Running   1 (117s ago)    7h4m
pod/user-service-57644fbdcf-zcdxw       0/1     Running   1 (118s ago)    7h4m

NAME                        TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
service/admin-service       ClusterIP   10.101.15.116    <none>        8007/TCP       7h48m
service/auth-service        ClusterIP   10.100.115.164   <none>        8001/TCP       7h48m
service/booking-service     ClusterIP   10.107.20.155    <none>        8004/TCP       7h48m
service/chargeops           NodePort    10.96.176.216    <none>        80:31573/TCP   7h51m
service/frontend            ClusterIP   10.111.124.225   <none>        80/TCP         7h48m
service/mongodb             ClusterIP   10.106.191.192   <none>        27017/TCP      7h48m
service/mongodb-headless    ClusterIP   None             <none>        27017/TCP      7h48m
service/payment-service     ClusterIP   10.97.222.189    <none>        8005/TCP       7h48m
service/review-service      ClusterIP   10.106.95.36     <none>        8006/TCP       7h48m
service/station-service     ClusterIP   10.100.58.251    <none>        8003/TCP       7h48m
service/user-service        ClusterIP   10.106.134.224   <none>        8002/TCP       7h48m

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/admin-service       0/1     1            0           7h48m
deployment.apps/auth-service        0/1     1            0           7h48m
deployment.apps/booking-service     0/1     1            0           7h48m
deployment.apps/chargeops           1/1     1            1           7h51m
deployment.apps/frontend            1/1     1            1           7h48m
deployment.apps/payment-service     0/1     1            0           7h48m
deployment.apps/review-service      0/1     1            0           7h48m
deployment.apps/station-service     0/1     1            0           7h48m
deployment.apps/user-service        0/1     1            0           7h48m

NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/admin-service-5f8586984c      1         1         0       7h4m
replicaset.apps/admin-service-7687dfdfb5      0         0         0       7h48m
```

HOME PAGE - CHARGEOPS



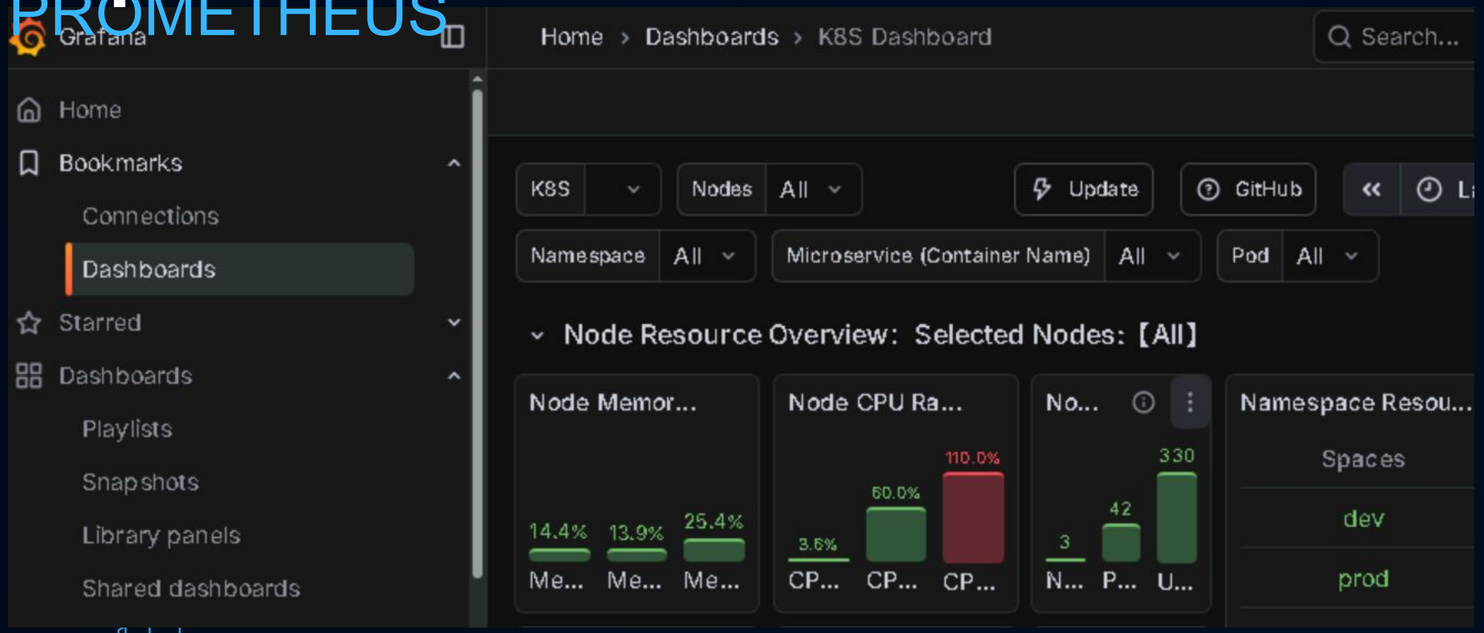
GRAFANA PROMETHEUS

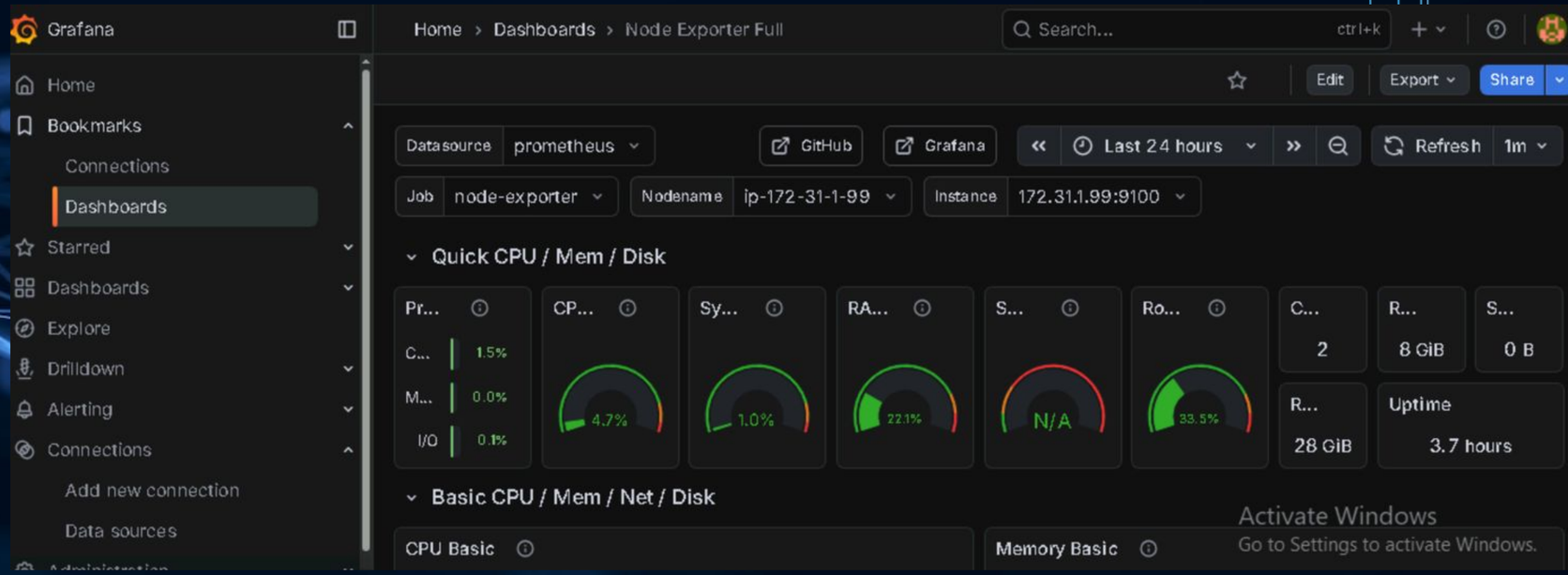




MONITORING DASHBOARD

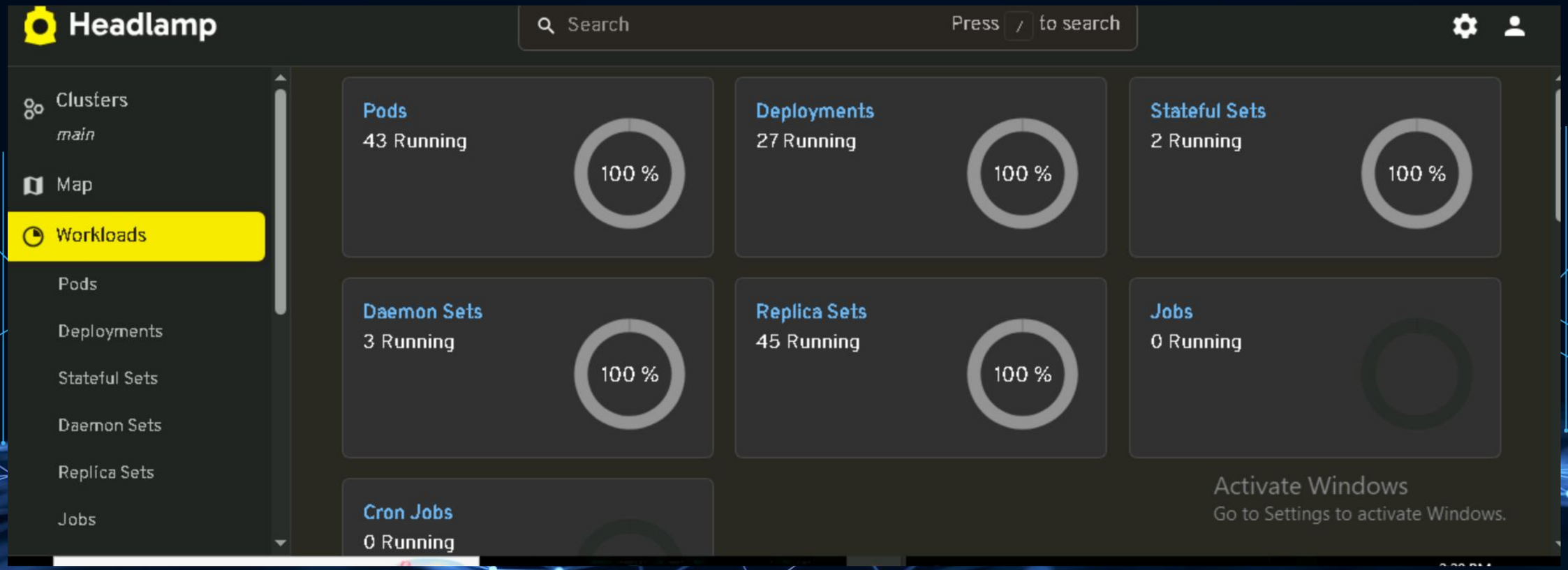
GRAFANA PROMETHEUS





MONITORING DASHBOARD

HEADLAMP


Headlamp
Search
Press / to search
Clusters
main
Map
Workloads
Pods
Deployments
Stateful Sets
Daemon Sets
Replica Sets
Jobs
Pods
43 Running
100 %
Deployments
27 Running
100 %
Stateful Sets
2 Running
100 %
Daemon Sets
3 Running
100 %
Replica Sets
45 Running
100 %
Jobs
0 Running
Cron Jobs
0 Running
Activate Windows
Go to Settings to activate Windows.

CONCLUSION

This project demonstrates a complete DevSecOps pipeline for a two-tier application.

It integrates Docker for containerization and Kubernetes for scalable deployment.

GitHub Actions automates build, test, security scanning, and deployment processes.

Security is ensured using SAST and container vulnerability scanning.

Separate dev and prod environments provide proper isolation and reliability.

Overall, it reflects real-world practices for building secure and efficient applications.